

A trade-off among complexity, performance, and space utilization determines the blocking strategy to be used.

A key function of any file management scheme is the management of disk space. Part of this function is the strategy for allocating disk blocks to a file. A variety of methods have been employed, and a variety of data structures have been used to keep track of the allocation for each file. In addition, the space on disk that has not been allocated must be managed. This latter function primarily consists of maintaining a disk allocation table indicating which blocks are free.

12.13 RECOMMENDED READING

[VENU09] presents an object-oriented design approach toward file structure implementation. [COME79] provides a thorough discussion of B-trees. [CORM09] and [KNUT98] also include good treatments.

The Linux file system is examined in detail in [LOVE10] and [BOVE06]. A good overview is [RUBI97].

BOVE06 Bovet, D., and Cesati, M. *Understanding the Linux Kernel*. Sebastopol, CA: O'Reilly, 2006.

COME79 Comer, D. "The Ubiquitous B-Tree." *Computing Surveys*, June 1979.

CORM09 Cormen, T., et al. *Introduction to Algorithms*. Cambridge, MA: MIT Press, 2009.

KNUT98 Knuth, D. *The Art of Computer Programming, Volume 3: Sorting and Searching*. Reading, MA: Addison-Wesley, 1998.

LOVE10 Love, R. *Linux Kernel Development*. Upper Saddle River, NJ: Addison-Wesley, 2010.

RUBI97 Rubini, A. "The Virtual File System in Linux." *Linux Journal*, May 1997.

VENU09 Venugopal, K. *File Structures Using C++*. New York: McGraw-Hill, 2009.

12.14 KEY TERMS, REVIEW QUESTIONS, AND PROBLEMS

Key Terms

access method	file	indexed sequential file
bit table	file allocation	inode
block	file allocation table	key field
chained file allocation	file directory	logical I/O
contiguous file	file management	pathname
allocation	system	physical I/O
database	file name	pile
device driver	hashed file	record
disk allocation table	indexed file	sequential file
field	indexed file allocation	working directory

Review Questions

- 12.1. What is the difference between a field and a record?
- 12.2. What is the difference between a file and a database?
- 12.3. What is a file management system?
- 12.4. What criteria are important in choosing a file organization?
- 12.5. List and briefly define five file organizations.
- 12.6. Why is the average search time to find a record in a file less for an indexed sequential file than for a sequential file?
- 12.7. What are typical operations that may be performed on a directory?
- 12.8. What is the relationship between a pathname and a working directory?
- 12.9. What are typical access rights that may be granted or denied to a particular user for a particular file?
- 12.10. List and briefly define three blocking methods.
- 12.11. List and briefly define three file allocation methods.

Problems

- 12.1. Define:
 - B = block size
 - R = record size
 - P = size of block pointer
 - F = blocking factor; expected number of records within a block
 Give a formula for F for the three blocking methods depicted in Figure 12.8.
- 12.2. One scheme to avoid the problem of preallocation versus waste or lack of contiguity is to allocate portions of increasing size as the file grows. For example, begin with a portion size of one block, and double the portion size for each allocation. Consider a file of n records with a blocking factor of F , and suppose that a simple one-level index is used as a file allocation table.
 - a. Give an upper limit on the number of entries in the file allocation table as a function of F and n .
 - b. What is the maximum amount of the allocated file space that is unused at any time?
- 12.3. What file organization would you choose to maximize efficiency in terms of speed of access, use of storage space, and ease of updating (adding/deleting/modifying) when the data are
 - a. updated infrequently and accessed frequently in random order?
 - b. updated frequently and accessed in its entirety relatively frequently?
 - c. updated frequently and accessed frequently in random order?
- 12.4. For the B-tree in Figure 12.4c, show the result of inserting the key 97.
- 12.5. An alternative algorithm for insertion into a B-tree is the following: As the insertion algorithm travels down the tree, each full node that is encountered is immediately split, even though it may turn out that the split was unnecessary.
 - a. What is the advantage of this technique?
 - b. What are the disadvantages?
- 12.6. Both the search and the insertion time for a B-tree are a function of the height of the tree. We would like to develop a measure of the worst-case search or insertion time. Consider a B-tree of degree d that contains a total of n keys. Develop an inequality that shows an upper bound on the height h of the tree as a function of d and n .

- 12.7.** Ignoring overhead for directories and file descriptors, consider a file system in which files are stored in blocks of 16K bytes. For each of the following file sizes, calculate the percentage of wasted file space due to incomplete filling of the last block: 41,600 bytes; 640,000 bytes; 4,064,000 bytes.
- 12.8.** What are the advantages of using directories?
- 12.9.** Directories can be implemented either as “special files” that can only be accessed in limited ways or as ordinary data files. What are the advantages and disadvantages of each approach?
- 12.10.** Some operating systems have a tree-structured file system but limit the depth of the tree to some small number of levels. What effect does this limit have on users? How does this simplify file system design (if it does)?
- 12.11.** Consider a hierarchical file system in which free disk space is kept in a free space list.
- a.** Suppose the pointer to free space is lost. Can the system reconstruct the free space list?
 - b.** Suggest a scheme to ensure that the pointer is never lost as a result of a single memory failure.
- 12.12.** In UNIX System V, the length of a block is 1 Kbyte, and each block can hold a total of 256 block addresses. Using the inode scheme, what is the maximum size of a file?
- 12.13.** Consider the organization of a UNIX file as represented by the inode (Figure 12.15). Assume that there are 12 direct block pointers, and a singly, doubly, and triply indirect pointer in each inode. Further, assume that the system block size and the disk sector size are both 8K. If the disk block pointer is 32 bits, with 8 bits to identify the physical disk and 24 bits to identify the physical block, then
- a.** What is the maximum file size supported by this system?
 - b.** What is the maximum file system partition supported by this system?
 - c.** Assuming no information other than that the file inode is already in main memory, how many disk accesses are required to access the byte in position 13,423,956?